Aurélien Cécille

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Education

Ph.D. Candidate in Computer Science

LIRIS & Visual Behavior — *May 2023 - June 2026*

Structure Aware Self-Supervised Monocular Depth Estimation

Master's Degree in Computer Science

University of Technology of Troyes — *2017 - 2022*

Work Experience

Research Engineer

Visual Behavior — *Oct. 2022 - Oct. 2025*

- Developed computer vision modules for intelligent vehicles and mobile robotics
- Estimated lead-vehicle distance and detected traffic lights for energy-aware driving applications
- Built local occupancy maps to improve robot environment awareness
- Contributed to ADAS perception for free-space detection around trucks and buses and vulnerable road user detection

AI Engineer Intern

Orange — *Feb. 2022 - Jul. 2022*

- Detection and segmentation of fiber optic cables in images
- Few-shot learning methods based on meta-learning
- Leveraging latent representations from generative models (GANs)

AI Engineer Intern

Visual Behavior — *Sept. 2020 - Feb. 2021*

- Self-supervised learning for depth estimation, optical flow, and egomotion estimation
- Environment reconstruction from bird's-eye view

Projects

Conditional Logo Generation

- Logo generation using StyleGAN2
- Conditioning the generator with latent-space clustering from a VAE
- Flask interface with color, shape, and category controls

Reinforcement Learning Agent

- Custom grid-world environment in Unity ML-Agents linked with OpenAI Gym
- Implementation in JAX for parallelization and optimized hyperparameter search

Research Interests

Computer Vision · Deep Learning · Self-Supervised Learning · Robotics

Publications

Towards Sharper Object Boundaries in Self-Supervised Depth Estimation

Aurélien Cécille, Stefan Duffner, Franck Davoine, Rémi Agier & Thibault Neveu

BMVC Oral, November 27, 2025, Sheffield, United Kingdom

GroCo: Ground Constraint for Metric Self-Supervised Monocular Depth

Aurélien Cécille, Stefan Duffner, Franck Davoine, Thibault Neveu & Rémi Agier

ECCV 2024, October 2024, Milan, Italy

Summer Schools & Workshops

ICVSS 2025

Computer Vision for Spatial Intelligence

Summer school attendance & poster

PAISS 2025

Summer school attendance

Naver Labs Workshop 2025

AI for Robotics

Attendance and poster presentation

Technical Skills

Core tools

PyTorch · JAX · Python · Docker

Languages

English : Fluent (C2 level)

French : Native language